H524 Introduction to Biostatistics

Week 5: Hypothesis Testing

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Outline

Quick Review: Confidence Intervals

Last week we covered:

- Point estimates vs interval estimates
- Confidence intervals for population mean
- General form: $\bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$
- Interpretation: 95% of intervals will contain μ
- Trade-off between confidence level and precision

This Week: Hypothesis testing - making decisions about population parameters

Reading: Pagano & Gauvreau Chapter 10

From Estimation to Testing

Two approaches to statistical inference:

1. Estimation (Weeks 3-4):

- What is the value of the parameter?
- Construct confidence interval
- Example: "True mean cholesterol is between 195 and 205 mg/dL"

2. Hypothesis Testing (This week):

- Is a specific claim about the parameter true?
- Make a decision: reject or fail to reject
- Example: "Is the true mean cholesterol greater than 200 mg/dL?"

Both use: Sample data, sampling distributions, probability theory

What is a Hypothesis Test?

Definition: A statistical procedure to assess evidence about a population parameter

Basic idea:

1. Make a claim about a population parameter
2. Collect sample data
3. Use probability to decide if data supports the claim
4. Make a decision: reject or fail to reject the claim

Medical example:

- Claim: "New drug lowers blood pressure by more than 10 mmHg on average"
- Collect data from clinical trial
- Calculate probability of observing this data if drug has no effect
- If probability is very small, reject "no effect" claim

The Logic of Hypothesis Testing

Key principle: *Proof by contradiction*

Analogy - Criminal trial:

- **Null hypothesis:** Defendant is innocent (default assumption)
- **Alternative hypothesis:** Defendant is guilty (what we're trying to prove)
- **Evidence:** Trial data
- **Decision:** Guilty (reject null) or Not Guilty (fail to reject null)
- **Standard:** Beyond reasonable doubt (low probability threshold)

Important: "Not guilty" $\neq$ "innocent"
Similarly: "Fail to reject H_0" $\neq$ "H_0 is true"

We never "accept" the null hypothesis - we only fail to reject it!

Null and Alternative Hypotheses

Null Hypothesis (H_0):

- Statement of "no effect" or "no difference"
- Status quo, default assumption
- What we assume is true unless proven otherwise
- Always contains equality: $=$, $\leq$, or $\geq$

Alternative Hypothesis (H_A or H_1):

- Statement of "effect exists" or "difference exists"
- What the researcher wants to demonstrate
- What we conclude if evidence is strong enough
- Contains inequality: $\neq$, $>$, or $<$

Key rule: H_0 and H_A must be:

- Mutually exclusive (can't both be true)
- Exhaustive (one must be true)

Types of Hypotheses: Two-Sided vs One-Sided

Two-sided (two-tailed) test:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_A : \mu \neq \mu_0$$

- Testing if parameter differs from null value (either direction)
- Example: "Is mean cholesterol different from 200 mg/dL?"
- Most common, more conservative

One-sided (one-tailed) test - Upper tail:

$$H_0 : \mu \leq \mu_0 \quad \text{vs} \quad H_A : \mu > \mu_0$$

- Testing if parameter is greater than null value
- Example: "Is mean blood pressure higher than 120 mmHg?"

One-sided test - Lower tail:

$$H_0 : \mu \geq \mu_0 \quad \text{vs} \quad H_A : \mu < \mu_0$$

- Testing if parameter is less than null value
- Example: "Does drug reduce cholesterol below 200 mg/dL?"

Example: Setting Up Hypotheses

Scenario 1: Normal body temperature is claimed to be 98.6°F. We want to test if this is accurate.

$$H_0 : \mu = 98.6 \quad \text{vs} \quad H_A : \mu \neq 98.6$$

Scenario 2: A new drug claims to lower blood pressure. Current mean is 140 mmHg.

$$H_0 : \mu \geq 140 \quad \text{vs} \quad H_A : \mu < 140$$

Scenario 3: We suspect exposure to a toxin increases cancer rate above baseline 5%.

$$H_0 : p \leq 0.05 \quad \text{vs} \quad H_A : p > 0.05$$

Key decision: Choose one-sided vs two-sided *before* collecting data, based on research question!

The Test Statistic

Definition: A value calculated from sample data that measures how far the sample is from the null hypothesis

General form:

$$\text{Test Statistic} = \frac{\text{Estimate} - \text{Null Value}}{\text{Standard Error}}$$

For testing mean (large sample or σ known):

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

For testing mean (σ unknown, small sample):

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where μ_0 is the null hypothesis value

What does the test statistic tell us?

- Measures "how many standard errors away" the sample mean is from null value
- Large $|t|$ or $|Z|$: Sample mean far from null value
- Small $|t|$ or $|Z|$: Sample mean close to null value

Examples:

- $t = 0.5$: Sample mean is 0.5 SE from null (not unusual)
- $t = 2.5$: Sample mean is 2.5 SE from null (unusual!)
- $t = -3.8$: Sample mean is 3.8 SE below null (very unusual!)

Question: How large does $|t|$ need to be for us to reject H_0?

Answer: Depends on:

- Degrees of freedom (sample size)
- Significance level (how much risk we'll accept)
- One-sided vs two-sided test

The P-value

Definition: The probability of observing a test statistic as extreme as (or more extreme than) what we observed, *assuming H_0 is true*

For two-sided test:

$$p\text{-value} = \Pr(|T| \geq |t_{obs}| \mid H_0 \text{ true})$$

For one-sided test (upper tail):

$$p\text{-value} = \Pr(T \geq t_{obs} \mid H_0 \text{ true})$$

For one-sided test (lower tail):

$$p\text{-value} = \Pr(T \leq t_{obs} \mid H_0 \text{ true})$$

Interpretation:

- Small p-value: Observed data would be very unlikely if H_0 were true
- Large p-value: Observed data consistent with H_0

Understanding P-values

Common misconceptions:

- $\times$ p-value is NOT $\Pr(H_0$ is true)
- $\times$ p-value is NOT Pr(data due to chance)
- $\times$ 1 - p-value is NOT $\Pr(H_A$ is true)

Correct interpretation:

"If the null hypothesis were true, the probability of obtaining a test statistic as extreme as or more extreme than the one we observed is [p-value]"

Rule of thumb for interpretation:

- $p < 0.01$: Very strong evidence against H_0
- $0.01 \leq p < 0.05$: Strong evidence against H_0
- $0.05 \leq p < 0.10$: Weak evidence against H_0
- $p \geq 0.10$: Little or no evidence against H_0

Significance Level (α)

Definition: The probability threshold for rejecting H_0

Decision rule:

- If p-value $< \alpha$: Reject H_0 (statistically significant)
- If p-value $\geq \alpha$: Fail to reject H_0 (not statistically significant)

Common choices:

- $\alpha = 0.05$ (most common in biostatistics)
- $\alpha = 0.01$ (more stringent, reduces false positives)
- $\alpha = 0.10$ (more lenient, exploratory studies)

Important: Choose α *before* collecting data!

Changing α after seeing results is called "p-hacking" and invalidates the test

Two Types of Errors

Hypothesis testing is a decision with uncertainty

		Reality (Unknown) H_0 True	H_0 False
Decision	Reject H_0	Type I Error (α)	Correct (Power)
	Fail to Reject	Correct ($1-\alpha$)	Type II Error (β)

Type I Error: Reject H_0 when it's actually true (false positive)

- Probability $= \alpha$ (significance level)
- Example: Conclude drug works when it doesn't

Type II Error: Fail to reject H_0 when it's actually false (false negative)

- Probability $= \beta$
- Example: Miss a real drug effect

Type I and Type II Errors: Medical Examples

Example 1: Disease screening test

H_0: Patient does not have disease
H_A: Patient has disease

- **Type I Error:** Diagnose disease when patient is healthy (false positive)
  - Consequence: Unnecessary treatment, anxiety
- **Type II Error:** Miss disease when patient is sick (false negative)
  - Consequence: Delayed treatment, disease progression

Example 2: Drug approval

H_0: Drug has no effect
H_A: Drug is effective

- **Type I Error:** Approve ineffective drug
  - Consequence: Patients receive useless treatment
- **Type II Error:** Reject effective drug
  - Consequence: Beneficial treatment unavailable

Controlling Type I Error

Type I Error rate = α (significance level)

We control Type I error by choosing α:

- Set $\alpha = 0.05$: 5% chance of false positive
- Set $\alpha = 0.01$: 1% chance of false positive (more conservative)
- Set $\alpha = 0.10$: 10% chance of false positive (more lenient)

Trade-off:

- Decreasing α (stricter) reduces Type I errors
- But increases Type II errors (β) - harder to detect real effects
- Can't make both arbitrarily small without increasing sample size

Convention in biostatistics:

- Typically prioritize controlling Type I error
- $\alpha = 0.05$ is standard (sometimes 0.01 for critical decisions)
- Better to miss a real effect than falsely claim one exists

Statistical Power

Power = Probability of correctly rejecting H_0 when H_A is true

$$\text{Power} = 1 - \beta$$

What affects power?

1. **Sample size (n):** Larger $n \Rightarrow$ higher power
2. **Effect size:** Larger true difference $\Rightarrow$ higher power
3. **Significance level (α):** Larger $\alpha \Rightarrow$ higher power
4. **Variability (σ):** Lower $\sigma \Rightarrow$ higher power

Typical goal: Power ≥ 0.80 (80%)

Means 80% chance of detecting the effect if it truly exists

Power analysis: Calculate required sample size to achieve desired power

- Done *before* collecting data
- Ensures study is adequately sized to detect meaningful effects

One-Sample t-Test

Used when:

- Testing hypothesis about a population mean μ
- Population SD (σ) is unknown
- Sample is random and approximately normal (or n large by CLT)

Hypotheses:

Two-sided: $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$
One-sided: $H_0 : \mu \leq \mu_0$ vs $H_A : \mu > \mu_0$ (or opposite)

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Under H_0: $t \sim t_{n-1}$ (t-distribution with $n - 1$ degrees of freedom)

Decision:

- Calculate p-value from t-distribution
- Reject H_0 if $p < \alpha$

Example: Body Temperature

Question: Is mean body temperature really 98.6°F?

Data: Sample of $n = 130$ healthy adults

- Sample mean: $\bar{x} = 98.25$°F
- Sample SD: $s = 0.73$°F

Step 1: Set up hypotheses

$$H_0 : \mu = 98.6 \quad \text{vs} \quad H_A : \mu \neq 98.6$$

Step 2: Calculate test statistic

$$t = \frac{98.25 - 98.6}{0.73/\sqrt{130}} = \frac{-0.35}{0.064} = -5.47$$

Step 3: Find p-value

From t-distribution with df $= 129$: $p < 0.0001$ (two-sided)

Step 4: Conclusion

$p < 0.05$, so reject H_0. Strong evidence that mean body temperature is not 98.6°F.

Example: Blood Pressure Drug

Question: Does new drug lower systolic BP below 140 mmHg?

Data: $n = 25$ patients after treatment

- Sample mean: $\bar{x} = 135$ mmHg
- Sample SD: $s = 12$ mmHg

Step 1: Set up hypotheses (one-sided)

$$H_0 : \mu \geq 140 \quad \text{vs} \quad H_A : \mu < 140$$

Step 2: Calculate test statistic

$$t = \frac{135 - 140}{12/\sqrt{25}} = \frac{-5}{2.4} = -2.08$$

Step 3: Find p-value (one-sided, lower tail)

From t_{24} distribution: $p = 0.024$

Step 4: Conclusion

$p = 0.024 < 0.05$, reject H_0. Evidence that drug lowers BP below 140 mmHg.

Critical Values vs P-values

Two equivalent approaches to hypothesis testing:

Method 1: P-value approach (modern, preferred)

1. Calculate test statistic t
2. Calculate p-value
3. Compare p-value to α
4. Reject H_0 if $p < \alpha$

Method 2: Critical value approach (classical)

1. Calculate test statistic t
2. Find critical value(s) from t-table for given α
3. Compare $|t|$ to critical value
4. Reject H_0 if $|t| > t_{critical}$ (two-sided)

Advantage of p-values:

- Provides exact strength of evidence
- Can compare to any α level
- Software gives p-values automatically

Critical Values for Two-Sided Tests

For two-sided test at $\alpha = 0.05$:

Reject H_0 if $|t| > t_{0.975,n-1}$

df	$t_{0.975}$	Interpretation
5	2.571	$|t| > 2.571$ is significant
10	2.228	$|t| > 2.228$ is significant
20	2.086	$|t| > 2.086$ is significant
30	2.042	$|t| > 2.042$ is significant
100	1.984	$|t| > 1.984$ is significant
∞	1.960	$|t| > 1.960$ is significant

Note: As df increases, t critical values approach $Z = 1.96$

For large samples ($n > 30$), can use normal approximation

One-Sample z-Test

Used when:

- Testing hypothesis about population mean μ
- Population SD (σ) is **known** (rare in practice!)
- OR sample size is very large ($n > 100$)

Test statistic:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Under H_0: $Z \sim N(0,1)$ (standard normal)

Critical values for $\alpha = 0.05$:

- Two-sided: $|Z| > 1.96$
- One-sided: $Z > 1.645$ (upper) or $Z < -1.645$ (lower)

In practice: Almost always use t-test instead

- Rarely know σ without knowing μ
- For large n, t-test and z-test give nearly identical results

Example: IQ Test with Known SD

Scenario: IQ tests are designed to have $\sigma = 15$ (known). We want to test if a special program increases mean IQ above 100.

Data: $n = 64$ students in program

- Sample mean: $\bar{x} = 103$
- Known SD: $\sigma = 15$

Hypotheses:

$$H_0 : \mu \leq 100 \quad \text{vs} \quad H_A : \mu > 100$$

Test statistic:

$$Z = \frac{103 - 100}{15/\sqrt{64}} = \frac{3}{1.875} = 1.60$$

P-value: $\Pr(Z > 1.60) = 0.055$

Conclusion: $p = 0.055 > 0.05$, fail to reject H_0. Marginal evidence that program increases IQ.

Relationship Between CIs and Hypothesis Tests

Key insight: Confidence intervals and hypothesis tests are closely related!

For two-sided test at level α:

Reject $H_0 : \mu = \mu_0$ if and only if the $(1 - \alpha) \times 100\%$ CI does NOT contain μ_0

Example: Blood pressure drug data

- $n = 25$, $\bar{x} = 135$ mmHg, $s = 12$ mmHg
- 95% CI: $135 \pm 2.064 \times \frac{12}{\sqrt{25}} = (130.0, 140.0)$
- Test $H_0 : \mu = 140$ vs $H_A : \mu \neq 140$
- Since 140 is *in* the CI, we would *fail to reject* H_0 (for two-sided test)

Note: This equivalence only holds for two-sided tests!

For one-sided tests, use one-sided confidence bounds

Using CIs for Multiple Comparisons

Advantage of confidence intervals:

Can simultaneously test multiple null values!

Example: Cholesterol drug trial

- Sample: $n = 50$, $\bar{x} = 185$ mg/dL, $s = 30$ mg/dL
- 95% CI: $(176.5, 193.5)$ mg/dL

We can immediately conclude:

- $\mu = 200$? Not in CI, reject H_0 ($p < 0.05$)
- $\mu = 190$? In CI, fail to reject H_0 ($p > 0.05$)
- $\mu = 180$? In CI, fail to reject H_0 ($p > 0.05$)
- $\mu = 170$? Not in CI, reject H_0 ($p < 0.05$)

Interpretation: Any value inside the CI is "plausible" given the data
Any value outside is "implausible" at the $\alpha = 0.05$ level

Statistical vs Clinical Significance

Statistical significance: $p < \alpha$
Clinical significance: Meaningful in practice

These are NOT the same!

Example 1: Large sample, small effect

- Drug lowers BP by 2 mmHg on average
- With $n = 10000$: $p < 0.001$ (highly significant)
- But 2 mmHg reduction may not be clinically meaningful

Example 2: Small sample, large effect

- Drug lowers BP by 15 mmHg on average
- With $n = 10$: $p = 0.08$ (not significant at $\alpha = 0.05$)
- But 15 mmHg reduction is clinically important!
- Need larger sample to demonstrate statistical significance

Always consider: Effect size, confidence interval width, clinical context

Effect of sample size on statistical tests:

Larger samples ($n \uparrow$):

- Smaller standard error: $\text{SE} = s/\sqrt{n}$
- Larger test statistics: $t = \frac{\bar{x} - \mu_0}{\text{SE}}$
- Smaller p-values (easier to reject H_0)
- Higher power to detect small effects
- Narrower confidence intervals

Implication:

- With large enough n, even tiny effects become "significant"
- Always report:
  - Effect size (e.g., difference in means)
  - Confidence interval
  - Clinical/practical importance
- Don't just report "$p < 0.05$"!

Common Mistakes in Hypothesis Testing

1. **Confusing "fail to reject" with "accept"**
   - $\times$ "We accept the null hypothesis"
   - ✓ "We fail to reject the null hypothesis"
   - Absence of evidence $\neq$ evidence of absence
2. **Choosing one-sided vs two-sided after seeing data**
   - Must decide before collecting data based on research question
3. **Changing α after seeing p-value**
   - "p $=$ 0.06, let's use $\alpha = 0.10$" is p-hacking!
4. **Interpreting p-value as** Pr(H_0 **true**)
   - P-value is Pr(data $\mid H_0$), not Pr($H_0 \mid$ data)
5. **Ignoring assumptions**
   - t-test assumes approximate normality (or large n)
   - Check with histogram, Q-Q plot, or normality tests
6. **Confusing statistical and practical significance**
   - Always consider effect size and clinical importance

Checking Assumptions

One-sample t-test assumes:

1. Random sample from population
2. Observations are independent
3. Population is approximately normal (or n large)

How to check normality:

1. Visual methods:

- Histogram: Should be roughly bell-shaped
- Q-Q plot: Points should follow straight line
- Boxplot: Should be roughly symmetric

2. Formal tests:

- Shapiro-Wilk test: Tests H_0: data is normal
- Anderson-Darling test
- But: With large n, minor deviations become "significant"

3. Central Limit Theorem:

- For $n \geq 30$, t-test is robust to non-normality
- Exception: Extreme outliers or strong skewness

One-Sample t-Test in R

Example: Body temperature data

```
# Body temperature data
temps <- c(98.0, 98.2, 98.4, 98.6, 98.0, 97.8, 98.8,
           98.0, 98.4, 98.6, 98.5, 98.3)

# Descriptive statistics
mean(temps)          # 98.30
sd(temps)            # 0.311
length(temps)        # 12

# Two-sided test: H0: mu = 98.6 vs HA: mu != 98.6
t.test(temps, mu = 98.6, alternative = "two.sided")

# Output includes:
#   t-statistic, degrees of freedom, p-value
#   95% confidence interval, sample mean
```

One-Sided t-Test in R (Part 1 - Setup)

```r
# Blood pressure data (after treatment)
bp <- c(138, 142, 135, 140, 132, 128, 145, 136,
        139, 133, 141, 137, 134, 130, 143)

# One-sided test: H0: mu >= 140 vs HA: mu < 140
result <- t.test(bp, mu = 140, alternative = "less")

# Extract components
result$statistic    # t-statistic
result$parameter    # degrees of freedom
result$p.value      # p-value
result$conf.int     # one-sided confidence bound
```

One-Sided t-Test in R (Part 2 - Manual Calculation)

```
# Manual calculation
xbar <- mean(bp)
s <- sd(bp)
n <- length(bp)
t_stat <- (xbar - 140) / (s / sqrt(n))
p_value <- pt(t_stat, df = n-1)

cat("Sample mean:", round(xbar, 2), "\n")
cat("t-statistic:", round(t_stat, 3), "\n")
cat("p-value:", round(p_value, 4), "\n")
```

Checking Normality in R (Part 1 - Visual Checks)

```
# Generate some data
set.seed(524)
data <- rnorm(30, mean = 100, sd = 15)

# Visual checks
par(mfrow = c(1, 3))

# 1. Histogram
hist(data, breaks = 10, main = "Histogram",
     xlab = "Value", col = "lightblue")

# 2. Q-Q plot
qqnorm(data, main = "Q-Q Plot")
qqline(data, col = "red", lwd = 2)
```

Checking Normality in R (Part 2 - Tests)

```
# 3. Boxplot
boxplot(data, main = "Boxplot", ylab = "Value",
        col = "lightgreen")

par(mfrow = c(1, 1))

# Shapiro-Wilk test
shapiro.test(data)
# p > 0.05: fail to reject normality (data looks normal)
```

Power and Sample Size in R

```
# Calculate power for one-sample t-test
library(pwr)  # install.packages("pwr")

# Power for n = 30 (effect size d = 0.5)
pwr.t.test(n = 30, d = 5/10, sig.level = 0.05,
           type = "one.sample",
           alternative = "two.sided")
# Power approx 0.70

# Sample size needed for 80% power
pwr.t.test(d = 5/10, power = 0.80,
           sig.level = 0.05,
           type = "one.sample",
           alternative = "two.sided")
# n approx 34
```

Complete Hypothesis Test Example (Part 1 - Setup)

```r
# Research question: Is mean cholesterol > 200 mg/dL?
chol <- c(215, 198, 203, 225, 192, 210, 208, 195,
          218, 202, 212, 205, 220, 198, 207, 213,
          201, 209, 217, 204)

# Step 1: Descriptive statistics
cat("Sample size:", length(chol), "\n")
cat("Sample mean:", round(mean(chol), 2), "\n")
cat("Sample SD:", round(sd(chol), 2), "\n\n")

# Step 2: Check assumptions
shapiro.test(chol)  # Test normality

# Step 3: Conduct test (one-sided, upper tail)
result <- t.test(chol, mu = 200,
                 alternative = "greater",
                 conf.level = 0.95)
print(result)
```

Complete Hypothesis Test Example (Part 2 - Interpret)

```
# Step 4: Interpret
if(result$p.value < 0.05) {
  cat("\nReject H0. Evidence that mean cholesterol > 200.\n")
} else {
  cat("\nFail to reject H0. Insufficient evidence.\n")
}

# Step 5: Effect size
effect_size <- (mean(chol) - 200) / sd(chol)
cat("Cohen's d =", round(effect_size, 3), "\n")
```

AI for Hypothesis Testing

How AI can help this week:

- **Conceptual clarification:** Understanding p-values, Type I/II errors
- **Test selection:** Choosing appropriate test for your data
- **R code generation:** Conducting t-tests, checking assumptions
- **Interpretation:** Understanding output from `t.test()`
- **Power analysis:** Sample size calculations
- **Visualization:** Creating plots to check assumptions

Example prompts:

- "Explain the difference between Type I and Type II errors with a medical example"
- "Show R code to conduct one-sided t-test testing if mean ¿ 100"
- "How do I interpret a p-value of 0.03?"
- "Create Q-Q plot in R to check normality assumption"

AI Demo: Understanding P-values (Part 1)

Prompt: "Explain what a p-value of 0.02 means in plain language"

Claude AI Response:

```
A p-value of 0.02 means:

"If the null hypothesis were actually true, we would
expect to see a result as extreme as (or more extreme
than) what we observed only 2% of the time by random
chance alone."

In practical terms:
- This is considered statistically significant
  (p < 0.05)
- It provides evidence against the null hypothesis
- There's only a 2% chance of getting this result if
  the null hypothesis is true
- We would typically reject the null hypothesis
```

AI Demo: Understanding P-values (Part 2)

```
Important: This does NOT mean:
- There's a 2% chance the null hypothesis is true
- There's a 98% chance our result is "real"
- The effect is necessarily large or important

Always consider the effect size and clinical
significance in addition to the p-value.
```

AI Demo: Test Selection

Prompt: "I have data from 20 patients before and after treatment. What test should I use?"

Claude AI Response:

```
For before-and-after data on the SAME patients, you should use a
PAIRED t-test (not a one-sample t-test).

Here's why:
- You have two measurements per patient (before and after)
- The measurements are dependent (same person)
- You're interested in the change within individuals

R code:
before <- c(...)  # before measurements
after <- c(...)   # after measurements

# Method 1: Paired t-test directly
t.test(after, before, paired = TRUE)

# Method 2: Create difference variable
difference <- after - before
t.test(difference, mu = 0)  # Test if mean difference != 0

Both methods give identical results. The paired t-test is more
powerful than comparing groups independently because it accounts
for individual variation.
```

AI Demo: Checking Assumptions (Part 1 - Visual)

Prompt: "How do I check if my data is normal enough for a t-test?"
Claude AI Response:

```
# Check normality for t-test
data <- c(...)  # your data

# Visual checks (most important!)
par(mfrow = c(2, 2))

# 1. Histogram
hist(data, breaks = 10, main = "Histogram", prob = TRUE)
curve(dnorm(x, mean = mean(data), sd = sd(data)),
      add = TRUE, col = "red", lwd = 2)

# 2. Q-Q plot (most useful!)
qqnorm(data)
qqline(data, col = "red")
# Points should roughly follow the line

# 3. Boxplot
boxplot(data, main = "Boxplot")
# Should be roughly symmetric, check for outliers
```

AI Demo: Checking Assumptions (Part 2 - Tests)

```
# 4. Density plot
plot(density(data), main = "Density Plot")

# Formal test
shapiro.test(data)
# p > 0.05 suggests normality
# But with large n, minor deviations become "significant"

# Rule of thumb: t-test is robust for n >= 30 unless
# extreme outliers or strong skewness
```

AI Best Practices for Hypothesis Testing

DO:

- Ask AI to explain concepts
- Request code examples
- Verify calculations independently
- Check assumptions visually
- Ask for interpretation help
- Learn from AI explanations

DON'T:

- Blindly trust AI p-values
- Skip checking assumptions
- Use AI without understanding
- Choose test based on desired result
- Ignore effect sizes
- Forget clinical context

Critical thinking points:

1. Does the test AI suggested match your research question?
2. Are assumptions reasonable for your data?
3. Does the interpretation make practical sense?
4. Is statistical significance the same as clinical importance?

Summary: Hypothesis Testing Framework

Complete hypothesis testing procedure:

1. **State hypotheses:** H_0 vs H_A (before collecting data!)
2. **Choose significance level:** Usually $\alpha = 0.05$
3. **Check assumptions:** Normality, independence, random sampling
4. **Calculate test statistic:** $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$
5. **Find p-value:** From t-distribution with $n-1$ df
6. **Make decision:** Reject H_0 if $p < \alpha$
7. **State conclusion:** In context of research question
8. **Report effect size:** Difference, confidence interval, practical importance

Remember:

- Statistical significance $\neq$ practical importance
- Fail to reject $\neq$ accept
- Always consider Type I and Type II error consequences
- Report confidence intervals alongside p-values

Key Concepts to Remember

Hypotheses:

- H_0: null hypothesis (status quo, no effect)
- H_A: alternative hypothesis (what we want to show)
- Two-sided vs one-sided tests

Decision-making:

- P-value: probability of data or more extreme, given H_0 true
- α: significance level (Type I error rate)
- Reject H_0 if $p < \alpha$

Errors:

- Type I (α): False positive (reject true H_0)
- Type II (β): False negative (fail to reject false H_0)
- Power $= 1 - \beta$: Probability of detecting true effect

Tests covered:

- One-sample t-test: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
- One-sample z-test: $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (rare)

Next Week Preview

Week 6: Two-Sample Tests

We'll extend hypothesis testing to compare two groups:

- Two-sample t-test (independent groups)
- Paired t-test (dependent samples)
- Confidence intervals for difference in means
- Assumptions and diagnostics
- Effect sizes (Cohen's d)

Examples:

- Treatment vs control groups
- Before vs after measurements
- Drug A vs Drug B comparison

Reading: Chapter 11

Questions?

Thank you!

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